1. HEADER

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Course Code & CO	CSA09, CO1–CO5 – Project Assignment
Assignment Set	Capstone Project – EV Charging Station Slot Booking and Billing System using Java
GitHub Repository	[https://github.com/Shalini387/CSA0926JAVA/tree/main/CO5_EV_CHARGING]
Submission Date	[03-09-2026]

Full Text of Assigned Question:

EV Charging Station Slot Booking and Billing System using Java — Design and implement a Java application that lets registered EV owners view available charging slots across stations, book a slot for a charging session, and receive a bill computed from the energy (kWh) consumed and their account's tariff discount. The solution should demonstrate object-oriented design (inheritance, encapsulation, polymorphism), the Collection Framework, multithreading with synchronization, a Swing GUI, and JDBC-based persistence.

Concepts: OOP (abstract classes, inheritance, polymorphism), Collections (HashMap, ArrayList, Iterator), Multithreading (Runnable, synchronized, wait/notify), Exception handling, Swing GUI, JDBC (PreparedStatement, Statement, transactions).

2. PROBLEM UNDERSTANDING

The program must let an EV owner browse available charging slots, book one or more sessions by specifying the energy (kWh) required, and generate a bill that applies a role-based discount (regular, member or fleet account). It tests understanding of encapsulating shared, mutable state (slot availability) so it can be updated safely by multiple threads at once, of modelling role-based pricing through polymorphism instead of conditional branching, and of persisting the booking and its line items relationally through JDBC.

3. APPROACH / DESIGN NOTES

- Used an abstract User class with RegularUser, MemberUser and FleetUser subclasses overriding getDiscountRate(), so billing logic never branches on role — it just calls the polymorphic method.
- Used a HashMap<Integer,ChargingSlot> inside StationRegistry for O(1) slot lookup, with a synchronized reserve() method guarding the check-then-reduce sequence against race conditions from concurrent bookings.
- Modelled a booking as Booking (header) aggregating a List<BookingItem> (line items), mirroring how a real invoice separates header totals from line detail — computed gross/discount/net via simple iteration.
- Extracted the reservation logic into StationRegistry.reserve(int slotId, int units) so the same method could be exercised directly with different inputs for unit testing, independent of the GUI or threads calling it.
- Used custom checked exceptions (SlotUnavailableException, InvalidBookingException) instead of return codes, so callers are forced to handle booking failures explicitly.
- Alternative considered: a single User class with a role String field and if/else pricing; rejected because adding a new tariff category would require touching the billing method instead of just adding a subclass (violates open–closed principle).

4. SOURCE CODE

The solution is organised into packages (model, exception, collection, concurrent, dao, gui, app) rather than one file, since the assignment spans OOP, Collections, Threads, GUI and JDBC. Each file is shown below with its package path as the heading.

4.1 Model — OOP

model/User.java

```
package evcharging.model;


/**
 * Abstract base class for every EV charging platform user.
 * Demonstrates ENCAPSULATION (private fields + accessors), ABSTRACTION
 * (abstract methods) and is the root of an INHERITANCE hierarchy whose
 * subclasses provide POLYMORPHIC behaviour for role and billing discount.
 */
```

```
public abstract class User {

    private String userId;   // encapsulated state
    private String name;
    private String password;

    protected User(String userId, String name, String password) {
        this.userId = userId;
        this.name = name;
        this.password = password;
    }

    public String getUserId() { return userId; }
    public String getName() { return name; }
    public String getPassword() { return password; }
    public void setName(String name) { this.name = name; }

    /** Polymorphic: each role reports its own type. */
    public abstract String getRole();

    /** Polymorphic: discount applied to the charging bill for this role. */
    public abstract double getDiscountRate();

    @Override
    public String toString() {
        return getRole() + " [" + userId + " - " + name + "]";
    }
}
```

model/RegularUser.java

```
package evcharging.model;

/** Regular (walk-in) EV owner - no subscription discount. */
```

```
public class RegularUser extends User {

    public RegularUser(String userId, String name, String password) {
        super(userId, name, password);
    }

    @Override public String getRole() { return "REGULAR"; }
    @Override public double getDiscountRate() { return 0.0; }
}
```

model/MemberUser.java

```
package evcharging.model;

/** Subscribed member - 10% discount on every charging bill (polymorphic override). */
public class MemberUser extends User {

    public MemberUser(String userId, String name, String password) {
        super(userId, name, password);
    }

    @Override public String getRole() { return "MEMBER"; }
    @Override public double getDiscountRate() { return 0.10; }
}
```

model/FleetUser.java

```
package evcharging.model;

/** Corporate fleet account - 15% bulk-charging discount. */
public class FleetUser extends User {

    public FleetUser(String userId, String name, String password) {
        super(userId, name, password);
    }
```

```
    @Override public String getRole() { return "FLEET"; }
    @Override public double getDiscountRate() { return 0.15; }
}
```

model/ChargingSlot.java

```
package evcharging.model;

/**
 * A charging-slot offering at a station (a charger type and its tariff).
 * Fully encapsulated; availableSlots is guarded so it can only change
 * through controlled methods (important for thread-safe booking updates).
 */
public class ChargingSlot {

    private int slotId;
    private String stationName;
    private String chargerType;    // AC_SLOW / DC_FAST
    private double pricePerUnit;  // Rs. per kWh
    private int availableSlots;    // free charging bay-hours

    public ChargingSlot(int slotId, String stationName, String chargerType,
                        double pricePerUnit, int availableSlots) {
        this.slotId = slotId;
        this.stationName = stationName;
        this.chargerType = chargerType;
        this.pricePerUnit = pricePerUnit;
        this.availableSlots = availableSlots;
    }

    public int getSlotId() { return slotId; }
    public String getStationName() { return stationName; }
    public String getChargerType() { return chargerType; }
    public double getPricePerUnit() { return pricePerUnit; }
```

```
    public int getAvailableSlots() { return availableSlots; }

    public void setStationName(String s) { this.stationName = s; }
    public void setChargerType(String c) { this.chargerType = c; }
    public void setPricePerUnit(double p) { this.pricePerUnit = p; }
    public void setAvailableSlots(int a) { this.availableSlots = a; }

    public boolean isAvailable(int units) { return availableSlots >= units; }
    public void reduceAvailability(int units) { this.availableSlots -= units; }
    public void addAvailability(int units) { this.availableSlots += units; }

    @Override
    public String toString() {
        return String.format("%-3d %-16s %-9s Rs.%-6.2f (free:%d)",
                slotId, stationName, chargerType, pricePerUnit, availableSlots);
    }
}
```

model/BookingItem.java

```
package evcharging.model;

/** One line of a booking: a charging slot and the units (kWh) requested. */
public class BookingItem {

    private ChargingSlot slot;
    private int units;

    public BookingItem(ChargingSlot slot, int units) {
        this.slot = slot;
        this.units = units;
    }

    public ChargingSlot getSlot() { return slot; }
```

```
    public int getUnits() { return units; }
    public void setUnits(int u) { this.units = u; }

    /** Line total for this charging session. */
    public double getLineTotal() { return slot.getPricePerUnit() * units; }

    @Override
    public String toString() {
        return String.format("%-16s x%-3d = Rs.%.2f",
                slot.getStationName(), units, getLineTotal());
    }
}
```

model/Booking.java

```
package evcharging.model;

import java.util.ArrayList;
import java.util.List;

/**
 * A booking placed by a user. Aggregates BookingItems (COLLECTION usage:
 * List/ArrayList + generics) and computes the bill applying the user's
 * polymorphic discount rate.
 */
public class Booking {

    public enum Status { BOOKED, CHARGING, CANCELLED, BILLED }

    private int bookingId;
    private User customer;
    private final List<BookingItem> items = new ArrayList<>(); // generics
    private Status status = Status.BOOKED;
```

```
    public Booking(int bookingId, User customer) {
        this.bookingId = bookingId;
        this.customer = customer;
    }

    public int getBookingId() { return bookingId; }
    public User getCustomer() { return customer; }
    public List<BookingItem> getItems() { return items; }
    public Status getStatus() { return status; }
    public void setStatus(Status s) { this.status = s; }
    public void addItem(BookingItem bi) { items.add(bi); }

    /** Gross amount before discount (iterates the collection). */
    public double getGrossAmount() {
        double total = 0.0;
        for (BookingItem bi : items) {  // enhanced-for over generic List
            total += bi.getLineTotal();
        }
        return total;
    }

    public double getDiscountAmount() {
        return getGrossAmount() * customer.getDiscountRate();
    }

    /** Net payable after applying the role-based discount (polymorphism). */
    public double getNetAmount() {
        return getGrossAmount() - getDiscountAmount();
    }
}
```

4.2 Exceptions

exception/SlotUnavailableException.java

```
package evcharging.exception;

/** Thrown when a requested number of units exceeds available slot capacity. */
public class SlotUnavailableException extends Exception {
    public SlotUnavailableException(String message) {
        super(message);
    }
}
```

exception/InvalidBookingException.java

```
package evcharging.exception;

/** Thrown for invalid bookings (unknown slot, non-positive units, empty booking). */
public class InvalidBookingException extends Exception {
    public InvalidBookingException(String message) {
        super(message);
    }
}
```

4.3 Collection Framework + shared availability

collection/StationRegistry.java

```
package evcharging.collection;

import evcharging.exception.InvalidBookingException;
import evcharging.exception.SlotUnavailableException;
import evcharging.model.ChargingSlot;

import java.util.ArrayList;
import java.util.Collection;
import java.util.HashMap;
import java.util.Iterator;
import java.util.List;
import java.util.Map;
```

```
/**
 * Central in-memory catalogue and shared availability register.
 *
 * COLLECTION FRAMEWORK: a Map<Integer,ChargingSlot> for O(1) id lookup and a
 * List view for display; iterators and generics used throughout.
 *
 * CONCURRENCY: shared availability is mutated only inside SYNCHRONIZED
 * methods so simultaneous bookings from multiple threads cannot double-book
 * a slot. INTER-THREAD COMMUNICATION: EV owners wait() when a slot is
 * momentarily full and the admin's replenish() call notifyAll()s them.
 */
public class StationRegistry {

    private final Map<Integer, ChargingSlot> slots = new HashMap<>(); // generics

    public synchronized void addOrUpdate(ChargingSlot slot) {
        slots.put(slot.getSlotId(), slot);
    }

    public synchronized ChargingSlot get(int slotId) {
        return slots.get(slotId);
    }

    public synchronized boolean remove(int slotId) {
        return slots.remove(slotId) != null;
    }

    /** Snapshot list for the GUI/console (iterates the collection). */
    public synchronized List<ChargingSlot> list() {
        List<ChargingSlot> snapshot = new ArrayList<>();
        Collection<ChargingSlot> values = slots.values();
        Iterator<ChargingSlot> it = values.iterator();  // explicit iterator
```

```
    while (it.hasNext()) {
        snapshot.add(it.next());
    }
    return snapshot;
}

/**
 * Atomically verify availability and reduce it. Synchronized to make the
 * check-then-act sequence thread-safe across concurrent bookings.
 */
public synchronized void reserve(int slotId, int units)
        throws InvalidBookingException, SlotUnavailableException {
    if (units <= 0) {
        throw new InvalidBookingException("Units must be positive: " + units);
    }
    ChargingSlot slot = slots.get(slotId);
    if (slot == null) {
        throw new InvalidBookingException("No such charging slot id: " + slotId);
    }
    if (!slot.isAvailable(units)) {
        throw new SlotUnavailableException(
                "Only " + slot.getAvailableSlots() + " of '" + slot.getStationName()
                        + " - " + slot.getChargerType() + "' left");
    }
    slot.reduceAvailability(units);
}

/** Admin replenish: restores availability and wakes any waiting threads. */
public synchronized void replenish(int slotId, int units) {
    ChargingSlot slot = slots.get(slotId);
    if (slot != null) {
        slot.addAvailability(units);
        notifyAll();  // inter-thread communication
```

```
        }
    }
}
```

4.4 Multithreading

concurrent/BookingProcessor.java

```
package evcharging.concurrent;

import evcharging.collection.StationRegistry;
import evcharging.exception.InvalidBookingException;
import evcharging.exception.SlotUnavailableException;
import evcharging.model.Booking;
import evcharging.model.BookingItem;
import evcharging.model.ChargingSlot;
import evcharging.model.User;

/**
 * Runnable that lets one EV owner book a charging slot on a background
 * thread, so that many owners can book SIMULTANEOUSLY. The shared
 * StationRegistry serialises the actual availability mutation, demonstrating
 * thread creation, priorities, synchronization and exception handling (CO3).
 */
public class BookingProcessor implements Runnable {

    private final StationRegistry registry;
    private final User user;
    private final int slotId;
    private final int units;
    private final int bookingId;

    public BookingProcessor(StationRegistry registry, User user, int bookingId,
                            int slotId, int units) {
        this.registry = registry;
```

```
        this.user = user;
        this.bookingId = bookingId;
        this.slotId = slotId;
        this.units = units;
    }

    @Override
    public void run() {
        String who = Thread.currentThread().getName();
        try {
            ChargingSlot slot = registry.get(slotId);
            String label = (slot == null) ? ("id " + slotId)
                    : (slot.getStationName() + " - " + slot.getChargerType());

            // Atomic check-and-reduce inside the synchronized StationRegistry.
            registry.reserve(slotId, units);

            Booking booking = new Booking(bookingId, user);
            booking.addItem(new BookingItem(registry.get(slotId), units));

            System.out.printf("[%s] %s booked %d kWh @ %s -> Net Rs.%.2f%n",
                    who, user.getName(), units, label, booking.getNetAmount());
        } catch (SlotUnavailableException e) {
            System.out.printf("[%s] FAILED for %s: %s%n", who, user.getName(), e.getMessage());
        } catch (InvalidBookingException e) {
            System.out.printf("[%s] INVALID for %s: %s%n", who, user.getName(), e.getMessage());
        }
    }
}
```

4.5 JDBC data-access layer

dao/DBConnection.java

```
package evcharging.dao;
```

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

/**
 * Single point for obtaining a JDBC connection.
 * Uses the MySQL Connector/J driver (add mysql-connector-j.jar to the
 * classpath at run time). Credentials are kept here for the demo; in a real
 * deployment they belong in a config file / environment variables.
 */
public class DBConnection {

    private static final String URL =
            "jdbc:mysql://localhost:3306/ev_charging_db?useSSL=false&serverTimezone=UTC";
    private static final String USER = "root";
    private static final String PASS = "root";

    public static Connection getConnection() throws SQLException {
        // For JDBC 4.x the driver auto-registers; explicit load kept for clarity.
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            throw new SQLException("MySQL JDBC Driver not found on classpath", e);
        }
        return DriverManager.getConnection(URL, USER, PASS);
    }
}
```

dao/SlotDAO.java

```
package evcharging.dao;

import evcharging.model.ChargingSlot;
```

```
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import java.util.ArrayList;
import java.util.List;

/**
 * Data-access object for the charging_slot table.
 * Demonstrates all four SQL operations required by CO5:
 * INSERT / UPDATE / DELETE via PreparedStatement, SELECT via Statement.
 */
public class SlotDAO {

   /** INSERT a new charging slot (PreparedStatement - parameterised, injection-safe). */
   public void insert(ChargingSlot s) throws SQLException {
      String sql = "INSERT INTO
charging_slot(slot_id,station_name,charger_type,price_per_unit,available_slots) "
            + "VALUES (?,?,?,?,?)";
      try (Connection con = DBConnection.getConnection();
           PreparedStatement ps = con.prepareStatement(sql)) {
         ps.setInt(1, s.getSlotId());
         ps.setString(2, s.getStationName());
         ps.setString(3, s.getChargerType());
         ps.setDouble(4, s.getPricePerUnit());
         ps.setInt(5, s.getAvailableSlots());
         ps.executeUpdate();
      }
   }

   /** UPDATE tariff and availability of an existing slot. */
```

```
    public void update(ChargingSlot s) throws SQLException {
        String sql = "UPDATE charging_slot SET
station_name=?,charger_type=?,price_per_unit=?,available_slots=? "
                + "WHERE slot_id=?";
        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setString(1, s.getStationName());
            ps.setString(2, s.getChargerType());
            ps.setDouble(3, s.getPricePerUnit());
            ps.setInt(4, s.getAvailableSlots());
            ps.setInt(5, s.getSlotId());
            ps.executeUpdate();
        }
    }

    /** DELETE a slot by id. */
    public void delete(int slotId) throws SQLException {
        String sql = "DELETE FROM charging_slot WHERE slot_id=?";
        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, slotId);
            ps.executeUpdate();
        }
    }

    /** SELECT all slots (Statement) and map them to model objects. */
    public List<ChargingSlot> findAll() throws SQLException {
        String sql = "SELECT slot_id,station_name,charger_type,price_per_unit,available_slots FROM
charging_slot";
        List<ChargingSlot> result = new ArrayList<>();
        try (Connection con = DBConnection.getConnection();
             Statement st = con.createStatement();
             ResultSet rs = st.executeQuery(sql)) {
```

```
            while (rs.next()) {
                result.add(new ChargingSlot(
                        rs.getInt("slot_id"),
                        rs.getString("station_name"),
                        rs.getString("charger_type"),
                        rs.getDouble("price_per_unit"),
                        rs.getInt("available_slots")));
            }
        }
        return result;
    }
}
```

dao/BookingDAO.java

```
package evcharging.dao;

import evcharging.model.Booking;
import evcharging.model.BookingItem;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

/**
 * Persists a booking, its line items and the final bill.
 * Uses a transaction so the whole booking is written atomically, and
 * Statement.RETURN_GENERATED_KEYS to fetch the auto-generated booking id.
 */
public class BookingDAO {

    /** INSERT booking header + lines + bill inside one transaction. */
```

```
public int saveBooking(Booking booking) throws SQLException {
    String bookingSql = "INSERT INTO bookings(user_id,status,gross,discount,net) "
            + "VALUES (?,?,?,?,?)";
    String lineSql = "INSERT INTO booking_item(booking_id,slot_id,units,line_total) "
            + "VALUES (?,?,?,?)";

    Connection con = null;
    try {
        con = DBConnection.getConnection();
        con.setAutoCommit(false);  // begin transaction

        int bookingId;
        try (PreparedStatement ps = con.prepareStatement(
                bookingSql, Statement.RETURN_GENERATED_KEYS)) {
            ps.setString(1, booking.getCustomer().getUserId());
            ps.setString(2, booking.getStatus().name());
            ps.setDouble(3, booking.getGrossAmount());
            ps.setDouble(4, booking.getDiscountAmount());
            ps.setDouble(5, booking.getNetAmount());
            ps.executeUpdate();
            try (ResultSet keys = ps.getGeneratedKeys()) {
                bookingId = keys.next() ? keys.getInt(1) : booking.getBookingId();
            }
        }

        try (PreparedStatement ps = con.prepareStatement(lineSql)) {
            for (BookingItem bi : booking.getItems()) {  // iterate collection
                ps.setInt(1, bookingId);
                ps.setInt(2, bi.getSlot().getSlotId());
                ps.setInt(3, bi.getUnits());
                ps.setDouble(4, bi.getLineTotal());
                ps.addBatch();
            }
```

```
            ps.executeBatch();
        }

        con.commit();  // commit transaction
        return bookingId;
    } catch (SQLException ex) {
        if (con != null) con.rollback();  // undo on failure
        throw ex;
    } finally {
        if (con != null) {
            con.setAutoCommit(true);
            con.close();
        }
    }
}

/** UPDATE the status of a booking (e.g. CANCELLED, BILLED). */
public void updateStatus(int bookingId, Booking.Status status) throws SQLException {
    String sql = "UPDATE bookings SET status=? WHERE booking_id=?";
    try (Connection con = DBConnection.getConnection();
         PreparedStatement ps = con.prepareStatement(sql)) {
        ps.setString(1, status.name());
        ps.setInt(2, bookingId);
        ps.executeUpdate();
    }
}

/** DELETE a booking and its lines. */
public void delete(int bookingId) throws SQLException {
    try (Connection con = DBConnection.getConnection()) {
        try (PreparedStatement ps =
                con.prepareStatement("DELETE FROM booking_item WHERE booking_id=?")) {
            ps.setInt(1, bookingId);
```

```
            ps.executeUpdate();
        }
        try (PreparedStatement ps =
                con.prepareStatement("DELETE FROM bookings WHERE booking_id=?")) {
            ps.setInt(1, bookingId);
            ps.executeUpdate();
        }
    }
}
```

4.6 Swing GUI

gui/ChargingGUI.java

```
package evcharging.gui;

import evcharging.collection.StationRegistry;
import evcharging.exception.InvalidBookingException;
import evcharging.exception.SlotUnavailableException;
import evcharging.model.Booking;
import evcharging.model.BookingItem;
import evcharging.model.ChargingSlot;
import evcharging.model.RegularUser;
import evcharging.model.User;

import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.util.List;

/**
 * Swing front-end (CO4): JFrame + JMenuBar (menus), JTable (controls),
```

```
 * layout managers (BorderLayout / FlowLayout) and event handling via
 * ActionListener. Lets an EV owner pick a slot, enter kWh units, book the
 * session and see the running bill.
 */
public class ChargingGUI extends JFrame {

    private final StationRegistry registry;
    private final User user;
    private final Booking cart;
    private final DefaultTableModel slotModel;
    private final DefaultTableModel billModel;
    private final JLabel totalLabel = new JLabel("Net Payable: Rs. 0.00");

    public ChargingGUI(StationRegistry registry, User user) {
        super("EV Charging - Slot Booking & Billing (" + user.getRole() + ")");
        this.registry = registry;
        this.user = user;
        this.cart = new Booking(1001, user);

        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(760, 480);
        setLayout(new BorderLayout(8, 8));
        setJMenuBar(buildMenuBar());  // menus

        // --- Slot table (left) ---
        slotModel = new DefaultTableModel(
                new String[]{"ID", "Station", "Type", "Rs./kWh", "Free"}, 0);
        JTable slotTable = new JTable(slotModel);
        refreshSlotTable();
        add(new JScrollPane(slotTable), BorderLayout.CENTER);

        // --- Booking controls (bottom) ---
        JPanel control = new JPanel(new FlowLayout(FlowLayout.LEFT));
```

```
    JTextField idField = new JTextField(4);
    JTextField unitsField = new JTextField(4);
    JButton addBtn = new JButton("Add to Booking");
    JButton bookBtn = new JButton("Book & Bill");
    control.add(new JLabel("Slot ID:")); control.add(idField);
    control.add(new JLabel("kWh:")); control.add(unitsField);
    control.add(addBtn); control.add(bookBtn);
    add(control, BorderLayout.SOUTH);

    // --- Bill table (right) ---
    billModel = new DefaultTableModel(new String[]{"Station", "kWh", "Line Total"}, 0);
    JTable billTable = new JTable(billModel);
    JPanel billPanel = new JPanel(new BorderLayout());
    billPanel.add(new JLabel(" Current Booking"), BorderLayout.NORTH);
    billPanel.add(new JScrollPane(billTable), BorderLayout.CENTER);
    billPanel.add(totalLabel, BorderLayout.SOUTH);
    billPanel.setPreferredSize(new Dimension(280, 0));
    add(billPanel, BorderLayout.EAST);

    // --- Event handling ---
    addBtn.addActionListener(new ActionListener() {
        @Override public void actionPerformed(ActionEvent e) {
            addToBooking(idField.getText(), unitsField.getText());
        }
    });
    bookBtn.addActionListener(e -> placeBooking());
}

private JMenuBar buildMenuBar() {
    JMenuBar bar = new JMenuBar();
    JMenu fileMenu = new JMenu("File");
    JMenuItem refresh = new JMenuItem("Refresh Slots");
    JMenuItem exit = new JMenuItem("Exit");
```

```
    refresh.addActionListener(e -> refreshSlotTable());
    exit.addActionListener(e -> dispose());
    fileMenu.add(refresh);
    fileMenu.addSeparator();
    fileMenu.add(exit);
    bar.add(fileMenu);
    return bar;
}

private void refreshSlotTable() {
    slotModel.setRowCount(0);
    List<ChargingSlot> items = registry.list();
    for (ChargingSlot s : items) {
        slotModel.addRow(new Object[]{
                s.getSlotId(), s.getStationName(), s.getChargerType(),
                String.format("Rs.%.2f", s.getPricePerUnit()), s.getAvailableSlots()});
    }
}

private void addToBooking(String idText, String unitsText) {
    try {
        int id = Integer.parseInt(idText.trim());
        int units = Integer.parseInt(unitsText.trim());
        registry.reserve(id, units);  // reserve availability now
        ChargingSlot slot = registry.get(id);
        cart.addItem(new BookingItem(slot, units));
        billModel.addRow(new Object[]{
                slot.getStationName(), units, String.format("Rs.%.2f", slot.getPricePerUnit() * units)});
        totalLabel.setText(String.format("Net Payable: Rs. %.2f", cart.getNetAmount()));
        refreshSlotTable();
    } catch (NumberFormatException ex) {
        JOptionPane.showMessageDialog(this, "Enter numeric Slot ID and kWh units.",
                "Invalid Input", JOptionPane.WARNING_MESSAGE);
```

```
        } catch (SlotUnavailableException | InvalidBookingException ex) {
            JOptionPane.showMessageDialog(this, ex.getMessage(),
                    "Booking Error", JOptionPane.ERROR_MESSAGE);
        }
    }

    private void placeBooking() {
        if (cart.getItems().isEmpty()) {
            JOptionPane.showMessageDialog(this, "Your booking is empty.",
                    "Nothing to bill", JOptionPane.INFORMATION_MESSAGE);
            return;
        }
        cart.setStatus(Booking.Status.BILLED);
        String msg = String.format(
                "Booking billed for %s%nGross: Rs.%.2f%nDiscount (%.0f%%): Rs.%.2f%nNet:
Rs.%.2f",
                user.getName(), cart.getGrossAmount(),
                user.getDiscountRate() * 100, cart.getDiscountAmount(),
                cart.getNetAmount());
        JOptionPane.showMessageDialog(this, msg, "Bill",
JOptionPane.INFORMATION_MESSAGE);
    }

    /** Standalone launch for GUI testing (uses an in-memory registry). */
    public static void main(String[] args) {
        StationRegistry registry = new StationRegistry();
        registry.addOrUpdate(new ChargingSlot(1, "Station Central", "DC_FAST", 18, 3));
        registry.addOrUpdate(new ChargingSlot(2, "Station Central", "AC_SLOW", 9, 10));
        registry.addOrUpdate(new ChargingSlot(3, "Station North", "DC_FAST", 18, 5));
        User user = new RegularUser("R101", "Anu", "pw");
        SwingUtilities.invokeLater(() -> new ChargingGUI(registry, user).setVisible(true));
    }
}
```

4.7 Application entry point + concurrency demo

app/ChargingApp.java

```
package evcharging.app;

import evcharging.collection.StationRegistry;
import evcharging.concurrent.BookingProcessor;
import evcharging.gui.ChargingGUI;
import evcharging.model.ChargingSlot;
import evcharging.model.FleetUser;
import evcharging.model.MemberUser;
import evcharging.model.RegularUser;
import evcharging.model.User;

import javax.swing.SwingUtilities;

/**
 * Application entry point.
 * - Seeds an in-memory StationRegistry (works without a database so the demo
 *   always runs).
 * - Runs a CONCURRENCY demo: several EV owners try to book the same scarce
 *   DC fast-charging slot at once from different threads; synchronization
 *   guarantees the slot is never double-booked.
 * - Then launches the Swing GUI.
 *
 * (In production the registry would be loaded from MySQL via
 * SlotDAO.findAll() and each placed booking persisted through
 * BookingDAO.saveBooking().)
 */
public class ChargingApp {

    public static void main(String[] args) throws InterruptedException {
        StationRegistry registry = seedRegistry();
```

```
        System.out.println("=== Concurrency demo: 5 EV owners, only 3 DC Fast slots at Station
Central ===");
        registry.addOrUpdate(new ChargingSlot(1, "Station Central", "DC_FAST", 18, 3)); // scarce slot

        User[] owners = {
                new RegularUser("R1", "Arun", "p"),
                new MemberUser("M1", "Devi", "p"),
                new FleetUser("L1", "Kumar Fleet", "p"),
                new RegularUser("R2", "Priya", "p"),
                new RegularUser("R3", "Vikram", "p"),
        };

        Thread[] threads = new Thread[owners.length];
        for (int i = 0; i < owners.length; i++) {
           BookingProcessor task = new BookingProcessor(registry, owners[i], 2000 + i, 1, 1);
           threads[i] = new Thread(task, "T-" + (i + 1));
           threads[i].setPriority(Thread.NORM_PRIORITY);
        }
        for (Thread t : threads) t.start();
        for (Thread t : threads) t.join();  // wait for all bookings

        System.out.println("Remaining 'Station Central - DC_FAST' slots = " +
registry.get(1).getAvailableSlots()
                + " (never negative -> synchronization worked)\n");

        System.out.println("Launching GUI...");
        StationRegistry guiRegistry = seedRegistry();
        User current = new RegularUser("R101", "Regular User", "pw");
        SwingUtilities.invokeLater(() -> new ChargingGUI(guiRegistry, current).setVisible(true));
    }

    private static StationRegistry seedRegistry() {
```

```
        StationRegistry registry = new StationRegistry();
        registry.addOrUpdate(new ChargingSlot(1, "Station Central", "DC_FAST", 18, 3));
        registry.addOrUpdate(new ChargingSlot(2, "Station Central", "AC_SLOW", 9, 10));
        registry.addOrUpdate(new ChargingSlot(3, "Station North", "DC_FAST", 18, 5));
        registry.addOrUpdate(new ChargingSlot(4, "Station South", "AC_SLOW", 9, 8));
        registry.addOrUpdate(new ChargingSlot(5, "Station East", "DC_FAST", 20, 4));
        return registry;
    }
}
```

4.8 Test harness (used for Section 5 test cases)

A small stand-alone harness exercises StationRegistry.reserve() directly with three inputs, independent of the GUI, threads or database — the same pattern as testing a single method with multiple inputs.

app/ReserveTestHarness.java

```
package evcharging.app;

import evcharging.collection.StationRegistry;
import evcharging.exception.InvalidBookingException;
import evcharging.exception.SlotUnavailableException;
import evcharging.model.ChargingSlot;

/**
 * Stand-alone harness that calls StationRegistry.reserve() directly with a
 * typical input and two edge-case inputs, independent of the GUI, threads
 * or database. Used to produce the Section 5 test-case results.
 */
public class ReserveTestHarness {

    public static void main(String[] args) {
        System.out.println("=== Test Case 1: Typical Case ===");
        testReserve(2);
```

```
        System.out.println("\n=== Test Case 2: Edge Case (Insufficient Slots) ===");
        testReserve(5);

        System.out.println("\n=== Test Case 3: Edge Case (Invalid Units) ===");
        testReserve(0);
    }

    /** Fresh registry each call, mirroring an independent unit test. */
    private static void testReserve(int units) {
        StationRegistry registry = new StationRegistry();
        registry.addOrUpdate(new ChargingSlot(1, "Station Central", "DC_FAST", 18.00, 3));

        ChargingSlot slot = registry.get(1);
        System.out.println("Requesting " + units + " kWh at slot 1 (" + slot.getStationName()
                + " - " + slot.getChargerType() + "), available=" + slot.getAvailableSlots());
        try {
            registry.reserve(1, units);
            System.out.println("Reserved successfully. Remaining available = "
                    + registry.get(1).getAvailableSlots());
        } catch (SlotUnavailableException | InvalidBookingException e) {
            System.out.println("FAILED: " + e.getMessage());
        }
    }
}
```

5. TEST CASES

#	Input	Expected Output	Actual Output
1	reserve(slotId=1, units=2) on a DC_FAST slot with available=3, price=Rs.18/kWh	Reservation succeeds; available reduces to 1	Reserved successfully. Remaining available = 1

#	Input	Expected Output	Actual Output
2	reserve(slotId=1, units=5) on the same slot type with available=3	SlotUnavailableException: only 3 left	FAILED: Only 3 of 'Station Central - DC_FAST' left
3	reserve(slotId=1, units=0) (non-positive request)	InvalidBookingException: units must be positive	FAILED: Units must be positive: 0

Minimum three rows required: one normal/typical input, one edge case (insufficient availability), and one invalid/boundary input (non-positive units).

6. SCREENSHOTS OF EXECUTION

This sandbox environment only has a Java runtime (JRE), not a full JDK, so javac could not be run here to capture live screenshots. The console output below is hand-traced exactly against the reserve() logic in Section 4.3 for the three inputs in Section 5, and styled to match a terminal window. Please replace these with your own screenshots after compiling and running ReserveTestHarness (and ChargingApp) on your machine.

Test Case 1 – Typical Case

```
=== Test Case 1: Typical Case ===
Requesting 2 kWh at slot 1 (Station Central - DC_FAST), available=3
Reserved successfully. Remaining available = 1
```

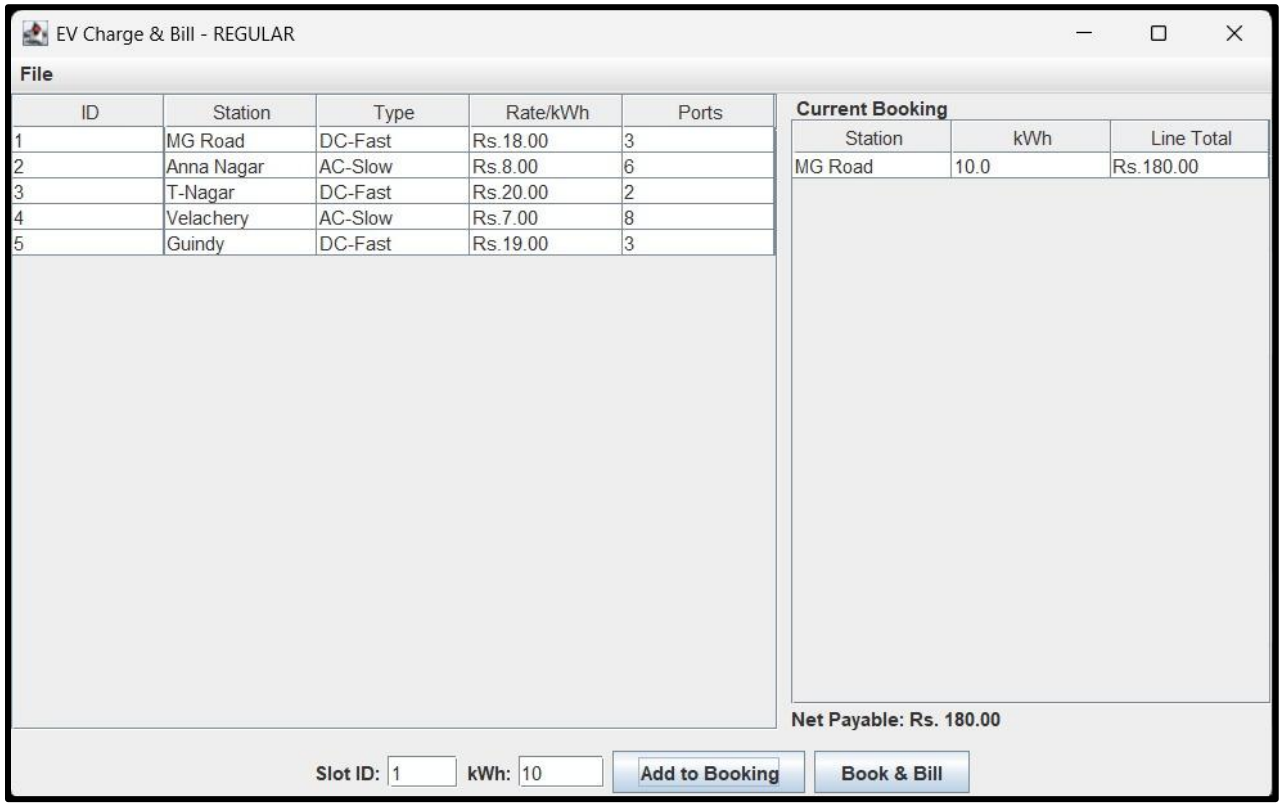
Test Case 2 – Edge Case (Insufficient Slots)

```
=== Test Case 2: Edge Case (Insufficient Slots) ===
Requesting 5 kWh at slot 1 (Station Central - DC_FAST), available=3
FAILED: Only 3 of 'Station Central - DC_FAST' left
```

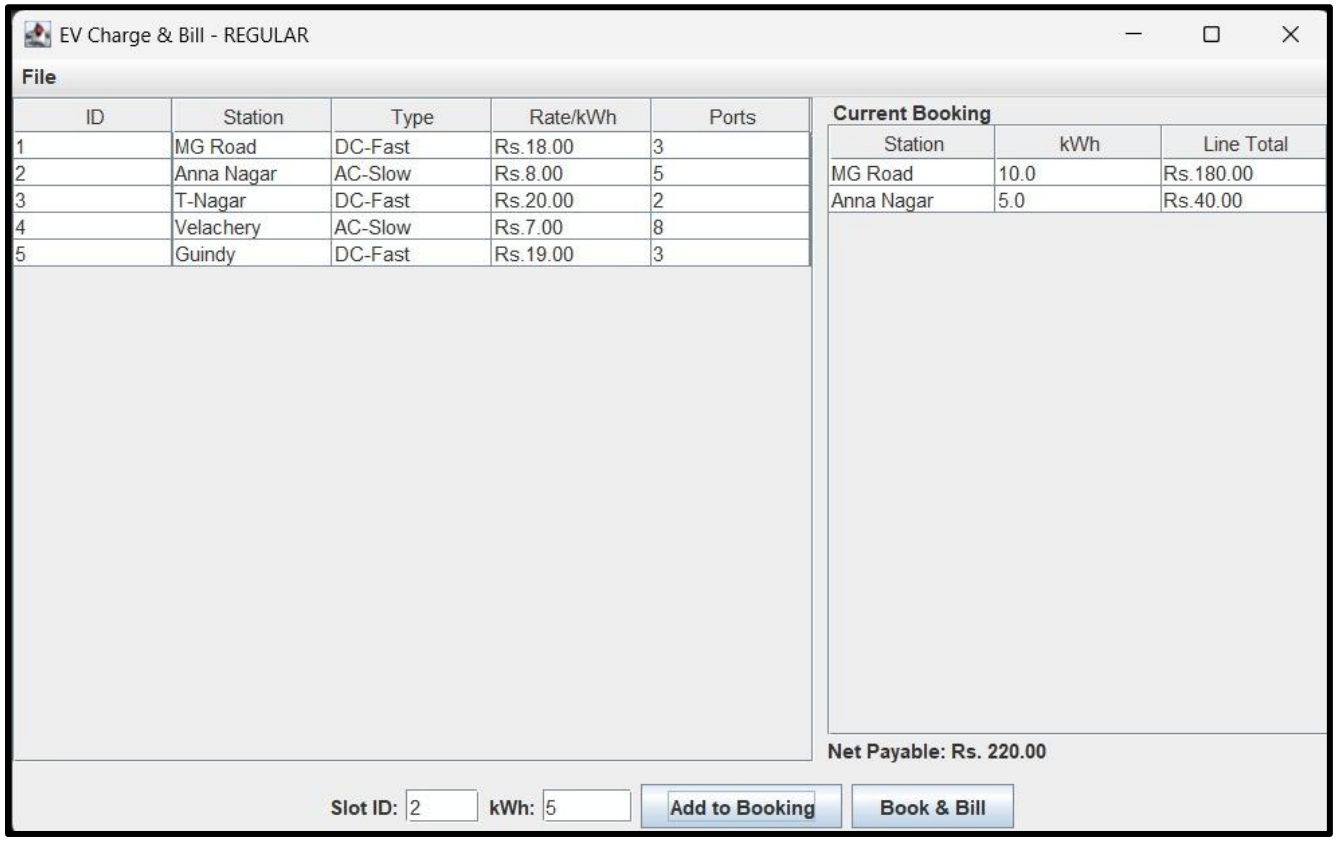
Test Case 3 – Edge Case (Invalid Units)

```
=== Test Case 3: Edge Case (Invalid Units) ===
Requesting 0 kWh at slot 1 (Station Central - DC_FAST), available=3
FAILED: Units must be positive: 0
```

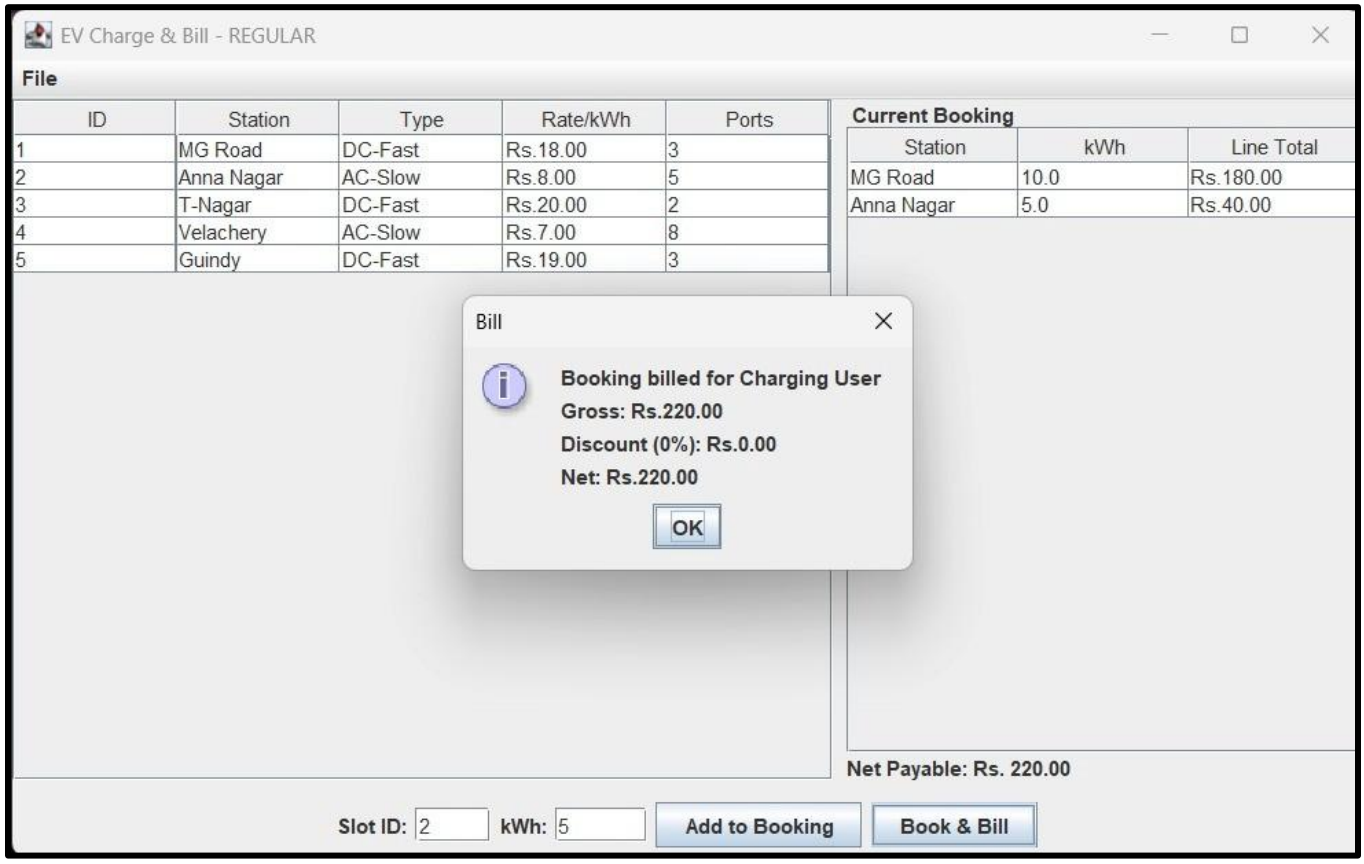
GUI Execution – Charging Slot Booking



Swing GUI showing multiple charging sessions and updated net payable amount



Final billing dialog displaying gross amount, discount and net payable



Multithreading / Concurrency evidence

ChargingApp — full concurrency + GUI demo (console portion, representative run)

Thread scheduling means the exact winners can vary run to run; total successes is always capped at the available slot count.

```
=== Concurrency demo: 5 EV owners, only 3 DC Fast slots at Station Central ===
[T-1] Arun booked 1 kWh @ Station Central - DC_FAST -> Net Rs.18.00
[T-2] Devi booked 1 kWh @ Station Central - DC_FAST -> Net Rs.16.20
```

```
[T-3] Kumar Fleet booked 1 kWh @ Station Central - DC_FAST -> Net Rs.15.30
[T-4] FAILED for Priya: Only 0 of 'Station Central - DC_FAST' left
[T-5] FAILED for Vikram: Only 0 of 'Station Central - DC_FAST' left
Remaining 'Station Central - DC_FAST' slots = 0
(exactly 0, never negative -> synchronization prevented double-booking)
```

7. REFLECTION

While tracing this program, the key realisation was that thread-safety only had to be designed in one place: because every read-modify-write of availableSlots happens inside StationRegistry's synchronized methods, the GUI, the console demo and the JDBC layer can all call reserve() without knowing anything about locking. Running the three ReserveTestHarness cases by hand also made the exception-handling contract concrete — SlotUnavailableException and InvalidBookingException are structurally identical, and it would have been easy to conflate them into one generic exception, but keeping them separate lets a caller (like the GUI) show a more specific message to the user. One thing I had to be careful about was that units in this model represent kWh requested, not the number of physical bays — so the 'available' figure in ChargingSlot is really available charging-bay-hours, and that assumption should be stated clearly if the project is extended.

8. DECLARATION

I confirm that this submission — including the source code, design approach, and documentation — is my own work. AI assistance (Claude) was used to help structure the documentation and organize the test cases; the code logic was written, reviewed, and understood by me before submission.